from Earth to spacecraft and through the wormhole so rapidly that he sees the Earth—spacecraft distance contracted from 10 light-years (10^{19} centimeters) down to the Planck—Wheeler length (10^{-33} centimeter). The laws of quantum gravity can take over and stop the growth of the beam only 10^{-43} second before the wormhole becomes a time machine, *as seen by such a circulating observer,* Hawking conjectured.

Translating back to the viewpoint of an observer at rest in the wormhole (the observer that Kim and I had relied on), Hawking's conjecture meant that the quantum gravity cutoff occurs 10^{-95} second before the wormhole becomes a time machine, not 10^{-43} second—and by then, according to our calculations, the vacuum fluctuational beam is strong enough, but just barely so, that *it might indeed destroy the wormhole.*

Hawking's conjecture about the location of the quantum gravity cutoff was cogent. He might well be right, Kim and I concluded after much contemplation; and we managed to change our paper to say so before it got published.

The bottom line, however, was equivocal. Even if Hawking was right, it was far from clear whether the beam of vacuum fluctuations would destroy the wormhole or not—and to find out for certain would require understanding what quantum gravity does, when it takes hold in the 10^{-95}-second interval around the moment of time machine formation.

To put it succinctly, *the laws if quantum gravity are hiding from us the answer to whether wormholes can be converted successfully into time machines.* To learn the answer, we humans must first become experts on quantum gravity's laws.

Hawking has a firm opinion on time machines. He thinks that nature abhors them, and he has embodied that abhorence in a conjecture, the *chronology protection* conjecture, which says that *the laws if physics do not allow time machines.* (Hawking, in his characteristic off-the-wall humor, describes this as a conjecture that will "keep the world safe for historians.")

Hawking suspects that the growing beam of vacuum fluctuations is nature's way of enforcing chronology protection: *Whenever one tries to make a time machine, and no matter what kind of device one uses in one's*